

Divya Bandaru

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EDUCATION

Cluster Innovation Centre, University of Delhi

2022 - present

Bachelor of Technology in Information Technology and Mathematics

CGPA: 9.09/10.0

Minor: Computational Biology

Coursework: In Silico Biology | Genes to Genome | Genomics & Proteomics | Linear Algebra | Calculus | Discrete Mathematics | Data Structure and Algorithms | Artificial Intelligence | Probability & Statistics | System Biology | Deep Learning

RESEARCH EXPERIENCE

National Centre for Biological Sciences, Bangalore

June 2024- Present

Advisor: Prof. Shaon Chakrabarti, Simons Centre for the Study of Living Machines, NCBS

- Developing an semi-supervised training pipeline using **Variational Autoencoder** to classify **cell images** into **cell cycle phases**.
- Extending the model to multimodal settings by testing VAE-based representations on tissue-scale spatial transcriptomics data (Xenium)

Graduate School of Science, University of Tokyo

June 2025 – August 2025

Amgen Scholars Program, Research Internship

Onsite (Tokyo, Japan)

Advisor: Prof. Yasushi Okada

- Studied dsRNA uptake and endosomal escape by **quantifying dose-dependent SID1-mediated uptake** and **distinguishing clathrin-mediated versus SID1-only pathways** using kinetic modelling.
- Conducted **quantitative and qualitative analysis** of **live-cell confocal fluorescence microscopy images** to study **colocalization** of **SLC proteins** with **lysosomes**.

International Centre for Genetic Engineering and Biotechnology (ICGEB), Delhi

Dec 2023 - April 2024

Advisor: Dr. Dinesh Gupta

- Developed a **DL-assisted diagnostic pipeline** for the **histopathological differential diagnosis** of **NIFTP** from different thyroid neoplasms, using **U-Net** and **EfficientNet**, achieving **87.6% accuracy**

PUBLICATION

- **A Novel Three-Stage AI-Assisted Approach for Accurate Differential Diagnosis and Classification of NIFTP and Thyroid Neoplasms**, *Endocrine Pathology*, 2025, <https://link.springer.com/article/10.1007/s12022-025-09865-0>

ACADEMIC PROJECTS

Multi-Omics Analysis of Aurora Kinase across 33 cancer types

- Integrated **TCGA** and **GTEX** RNA-seq, **HM450** DNA methylation, clinical survival data, and immune infiltration scores (ssGSEA).
- Applied **Mann–Whitney U**, **Kruskal–Wallis**, **Kaplan–Meier**, **Cox regression**, **ROC/Time-ROC**, and **nomogram** modeling.

Analysis of Genetic Variants Associated with Alzheimer’s Disease

- Performed **genetic association analysis** on **ADNI-3** data from **654 participants**, identifying Alzheimer’s-linked **SNPs** using **chi-square tests** and **logistic regression**, performing **ancestry comparisons** using **PCA**, **functional SNP annotation** using **FUMA**, and **gene-set pathway analysis** using **MAGMA**.

Detect PCOS from Ovarian Ultrasound Images and Statistical Data

- Developed a **CNN model** using **2,000 ovarian ultrasound images** to predict **PCOS** and created a **FastAPI-based API** to locally host and provide predictions from medical images.
- On the dataset of **541 patients** and **41 attributes**, addressed **class imbalance** with **SMOTE**, **SMOTE-ENN**, and **ADASYN (upsampling)** and **Random Undersampling** (downsampling), and evaluated classifiers like **Random Forest**, **SVM**, **KNN**, **Naive Bayes** and **Logistic Regression** on balanced datasets. **RFC** upsampled using **ADASYN** achieved **94.4% accuracy**.

SKILLS

Languages: Python, Julia, R, MATLAB

Tools: ImageJ, Ilastik, AutoDock Vina, GROMACS, PyMol, PyRx

Lab: Transfection, PCR, Timelapse Microscopy, UV/Vis Spectrophotometry, Fluorescence Spectroscopy